

Colour Profiles

A colour profile is a file that holds information about the colour reproduction capabilities of an imaging device such as a monitor or scanner. It is provided by the vendor, and is installed on the computer to enable the monitor to display colours correctly.

Among the 19-inch monitors, the Philips 109 B6 and the Samsung SyncMaster 957MB got the lines correct. The Samsung 957MB returned results as good as Philips. Both these 19-inch monitors will make good companions for those with eyes sensitive to correct geometry, as well as for CAD/CAM professionals.

The Pump Effect

When there is a sudden change in brightness, the screen expands or gets displaced. This is called the pump effect, and is a common problem with CRT technology. Vendors are yet to provide a solution for it.

You can see this effect by using a wallpaper with a dark background and running an office application with a light or white background—if you switch from the application to the wallpaper, you will see the screen contract. Switching back to the application will result in the screen expanding. This adversely affects your experience, especially while watching a movie.

In the 17-inch segment, the NEC MultiSync FE791SB returned solid results. There was very little pump effect visible. Some monitors such as the Samsung SyncMaster 793DF returned very poor results; the screen not only got bigger in size, but also got displaced by more than a millimetre. Therefore, if you are a movie buff, spend that extra money and get the NEC MultiSync FE791SB.

Amongst the 19-inch monitors, NEC MultiSync P912SB returned impressive results, followed by both the Samsung models. The Viewsonic UltraBright E90f+SB returned unacceptable results.

Resolution And Sharpness

Here, we were looking for sharpness in horizontal and vertical lines and bars, crispness of patterns in the middle and corners

of the screen, and the moiré problem. Sharpness suffers because monitors are not able to draw every detail correctly at the resolutions and refresh rates we usually work at. This results in serrated edges in bars and lines, both horizontal and vertical. Since the screen is drawn at very high speeds—typically 75 to 80 times a second—misconvergence occurs, which in turn results in a pattern appearing blurred at the corners.

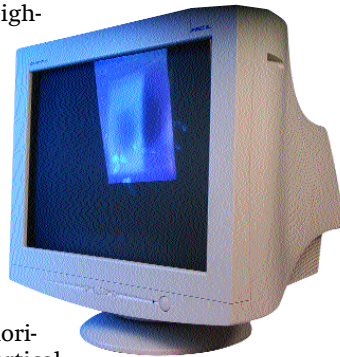
The other reason for imperfectly crisp images at the corners is the deflection of the electron beam from its path. The beam, which hits the middle of the screen, is not deflected much, and therefore doesn't lose its circular shape. However, when it hits the corners, the circular shape turns into an elliptical one, and as a result of this, neighbouring pixels get highlighted at the same time.

In this set of tests, the LG Studioworks 700S displayed crisp images at our test resolution of 1,024 x 768 at 75 Hz. It displayed sharp horizontal and vertical lines, and even single dots didn't bleed too much into neighbouring pixels. It also managed to display fine patterns generated by DisplayMate crisply in the middle and with very little fuzziness in the corners.

The Samsung 793S, an entry-level monitor, was not up to the mark. Horizontal lines appeared serrated, and the dots were nowhere near round. Even the patterns appeared rather hazy in the corners.

The moiré effect, which is another problem with CRT monitors—whether shadow mask-based or aperture-grille-based—happens not because the monitor is faulty, but because of the limitations imposed by the technology. A dot pitch as fine as 0.24 mm is not sufficient to eliminate moiré completely. Any finer dot or stripe pitch will increase the price of the monitor drastically, making it unfeasible for the mass market.

There is a moiré-correcting function on the On-Screen Display (OSD) controls of all monitors. But in most cases it doesn't help to eliminate moiré completely.



HCL HCM985RFM

The Acer AF715 and the Viewsonic E71f were the only monitors that provided a moiré-free display. There was absolutely no trace of line moiré, and the visible dot moiré was almost negligible. The LG Flatron E700SH and Samsung 793S returned images filled with moiré in both the line and dot tests. All attempts to correct this using the OSD were in vain.

In the 19-inch category, the HCL HCM985RFM returned the best results, with very little moiré in both the line and dot tests. The Philips 109 B6, too, returned commendable results, but the Philips 109 E5, on the other hand, had moiré smattered all over the screen. Even the NEC MultiSync FP912SB suffered in the dot moiré test.

Screen Pixel

This is a test that uses a curved and an inclined line. Since the dots in a CRT screen are arranged in triads, it is not possible to display an inclined line or curve smoothly. However, a monitor with a tighter dot pitch will serve up a less crooked line as compared to one with a coarse dot pitch.

There was hardly any difference among monitors which were tested in both the 17-inch and 19-inch categories.

Colour And Greyscale

Here, we looked for the purity of primary and secondary colours, reverse text, the ghosting effect, and level shift. Ghosting and streaking is an effect visible at the edges of the display area where the intensity of the brightness suddenly drops. This effect is lower when a CRT is new, but increases as it gets older.

This can be very irritating for those working on MS Word or a programme with a similar layout, where a bright area is adjacent to a dark area. Tweaking the contrast, in many cases, can take care of this problem to some extent, but it can cause strain while reading text.

What Is Colour Temperature?

A certain colour temperature is defined as the colour emitted by a black body at that particular temperature. Colour temperature, also known as White Point, is a setting in almost all monitors. It is adjusted so that the monitor displays a natural-looking colour. 9300 K is termed Cool White, and 6700 K is seen as a warmer white.